

FINAL PROJECT

End-to-End Statistical Analysis of a Complex Real-World Dataset

By: Dr Hamza Altarturi

Submission	Week 13 – exact date TBA	Group Size	4–5 students
Total Marks	100 (course weight: 40%)	Format	Notebook (code) + Technical Report (word doc) + Presentation (ppt slides)
Dataset	One per group – pre-assigned	Presentation	15 minutes + 5 min Q&A (Week 13)

1. Purpose and Scope

This project is the culminating assessment of the course. You will apply the full analytical pipeline developed across Weeks 1 through 13 to a large, complex, real-world dataset. The goal is not to demonstrate that you can run code – it is to demonstrate that you can think statistically: frame a question, challenge your assumptions, choose methods with justification, interpret results in context, and communicate findings with precision and honesty.

The project is deliberately open-ended. There is no single correct model or analysis path. You will be assessed on the quality of your reasoning, the depth of your justifications, and your ability to recognise and acknowledge the limits of your analysis.

The central question your project must answer

Each project dataset comes with a primary analytical question (stated in Section 3). Your entire analysis must serve that question. Every method you apply must be motivated by that question – not applied as a checklist. A project that uses many methods without connecting them to the central question will score poorly on integration.

2. Mark Allocation

The project is marked out of 100 and comprises six parts plus a presentation component.

#	Title	Key Skills Assessed	Marks
1	Dataset Audit & Exploratory Analysis	EDA, variable classification, missingness, visualisation (Wks 1–3)	15
2	Inferential Statistics	Estimation, CIs, hypothesis testing, effect size (Wks 4–7)	15
3	Regression Modelling	SLR, MLR, feature engineering, diagnostics (Wks 8–10)	20
4	Classification & Logistic Regression	Logistic regression, odds ratios, model evaluation (Wk 11)	15
5	Model Validation & Regularization	Cross-validation, Ridge/Lasso, bias-variance (Wk 12)	15
6	Critical Synthesis & Statistical Pitfalls	Simpson’s paradox, p-hacking, causation, limitations (Wk 13)	10
P	Presentation & Group Q&A	Clarity, depth under questioning, teamwork	10
	TOTAL		100

Marks are not awarded for volume

A well-reasoned 30-page report outscores a bloated 50-page one. Marks are awarded for depth of statistical reasoning, accuracy of interpretation, and honest acknowledgement of limitations – not for the number of plots or models produced.

3. Project Datasets

Each group is assigned one dataset. Datasets are intentionally complex: they contain missing values, distributional irregularities, multicollinearity, imbalanced classes, and multiple plausible analytical paths. This complexity is the point – real data is never clean.

Dataset assignment

Groups must confirm their assigned dataset with the instructor by end of Week 10. A group may request a specific dataset but assignment is final once confirmed.

Dataset A – MIMIC-III ICU Patient Outcomes (Healthcare Analytics)

Attribute	Detail
Source	PhysioNet MIMIC-III Clinical Database (publicly available subset via Kaggle: mimic-iii-clinical-database-demo)
Rows/Cols	~46,000 hospital admissions × 25+ variables (derived feature table)
Key Variables	Age, gender, ethnicity, admission type, primary ICD-9 diagnosis group, length of stay (LOS), number of ICU stays, SAPS-II severity score, first-day vitals (heart rate, BP, SpO2, temp), 28-day in-hospital mortality (binary), discharge disposition
Primary Question	What patient and admission characteristics best predict 28-day in-hospital mortality, and how does illness severity (SAPS-II score) interact with demographic variables in that prediction?
Difficulty Factors	Heavy right-skew in LOS and age; class imbalance in mortality (~11% positive class); multicollinearity among severity scores and vitals; Simpson's paradox risk across ethnic subgroups; meaningful missing data in SpO2 and blood pressure fields

Dataset B – American Community Survey PUMS (Economic Inequality)

Attribute	Detail
Source	US Census Bureau – ACS 1-Year PUMS 2022 (person-level microdata, freely downloadable from census.gov/programs-surveys/acs)
Rows/Cols	~3.2 million person records; select 15,000 working-age records × 30+ variables
Key Variables	State, age, sex, race/ethnicity, educational attainment, occupation code (SOC), industry code (NAICS), usual hours worked per week, weeks worked, total personal income, wage/salary income, health insurance coverage (binary), disability status (binary), citizenship
Primary Question	After controlling for education, occupation, industry, and hours worked, what is the residual wage gap attributable to gender and race/ethnicity, and does this gap vary across educational levels?
Difficulty Factors	Complex survey design requires careful subsetting; income is heavily right-skewed and requires log transformation; occupation and industry codes need grouping before use; strong multicollinearity between education and occupation; intersectionality effects (gender × race) require interaction terms; zero incomes require careful handling before log transformation

Dataset C – Global Terrorism Database (Security & Political Science)

Attribute	Detail
Source	START Center, University of Maryland – GTD (globalterrorismdb_0522dist.csv , free academic download at start.umd.edu/gtd)

Rows/Cols	~210,000 incidents (1970–2021) × 40+ variables; restrict to 2000–2021 for project (~100,000 rows)
Key Variables	Year, month, country, region, attack type, weapon type, target type, perpetrator group name, success indicator (binary), suicide attack (binary), number killed, number wounded, property damage value, claimed responsibility (binary), hostage/kidnapping indicator
Primary Question	What attack-level, contextual, and perpetrator characteristics predict whether a terrorist incident results in casualties above the median, and have these predictors changed between 2000–2010 and 2011–2021?
Difficulty Factors	Extreme right-skew and zero-inflation in casualty counts; very high class imbalance for some attack types; substantial missing data in weapon type and perpetrator fields; temporal autocorrelation and country-level clustering violate independence; requires careful operationalisation of the outcome variable

Dataset D – Stack Overflow Annual Developer Survey (Labour Markets & Technology)

Attribute	Detail
Source	Stack Overflow Developer Survey 2023 (survey.stackoverflow.co/datasets – free download, ~90,000 respondents)
Rows/Cols	~89,000 respondents × 80+ questions; select relevant 25 columns after preprocessing
Key Variables	Country, years of professional coding experience, employment type, education level, primary programming languages (multi-select – requires encoding), annual compensation (USD, self-reported), AI tool usage (binary + frequency), remote work status, job satisfaction score (ordinal 1–5), developer type (front-end / back-end / data / DevOps / etc.)
Primary Question	What factors – experience, education, language portfolio, AI tool adoption, and geography – are the strongest predictors of annual compensation among software developers, and does AI tool adoption mediate the relationship between experience and compensation?
Difficulty Factors	Extreme outliers and currency conversion issues in compensation; multi-select categorical variables require binarisation (one-hot per language); self-selection and voluntary response bias must be discussed; country of residence creates clustering and requires grouping by region; ordinal job satisfaction variable must be justified as continuous or kept ordinal

Dataset E – WHO + World Bank Merged Health & Development Panel (Global Health)

Attribute	Detail
Source	WHO Global Health Observatory API + World Bank WDI (merged by country-year); see data.worldbank.org and who.int/data/gho. Pre-merged CSV available on instructor’s course portal.
Rows/Cols	~4,200 country-year observations (180 countries × 2000–2022) × 28 variables
Key Variables	Country, year, region (WHO), income group (World Bank), population, under-5 mortality rate (U5MR), maternal mortality ratio, life expectancy at birth (by sex), adult obesity prevalence, adult smoking prevalence, HIV prevalence, health expenditure per capita (USD PPP), GDP per capita (USD PPP), physician density (per 1,000 population), safe water access (%), sanitation access (%), vaccination coverage (DTP3 %, measles %)
Primary Question	Across low- and middle-income countries, what combination of health system inputs (expenditure, physician density, vaccination) and socioeconomic conditions best predicts under-5 mortality rate, and is the effect of health expenditure modified by income group?
Difficulty Factors	Panel data structure violates independence assumption; strong multicollinearity between GDP, health expenditure, and physician density; U5MR is right-skewed and benefits from log transformation; missing data is not random (MNAR – poorest countries report least); requires group-level analysis to reveal Simpson’s paradox across income groups; temporal trends must be acknowledged even if not formally modelled

4. Deliverables

Your group must submit the following as a single compressed archive (.zip) via the course portal:

#	Deliverable	Format	Max Length
1	Jupyter Notebook	.ipynb – fully re-executable, relative paths, no hardcoded absolute paths	No limit on code; 300 words max per inline written response cell
2	Technical Report	PDF or Word. Structured by Part (1–6). Figures may be embedded.	35 pages (excluding references and appendices)
3	Presentation Slides	PDF or PowerPoint. One deck per group.	20 slides maximum
4	Group Contribution Statement	Plain text or PDF. One paragraph per member stating specific contributions.	1 page
5	Processed Data File(s)	CSV or Parquet. Final cleaned dataset as used in modelling.	As required

Critical: notebook reproducibility

The notebook will be re-run by the marker from top to bottom. Any notebook that fails to execute, produces different results on re-run, or contains broken file paths will be marked as non-reproducible and capped at 50% of the notebook-dependent marks.

5. Project Questions

The following questions apply to all groups regardless of dataset. Dataset-specific guidance is provided in italics where the question requires it. The questions are ordered by Part and build on each other – your answers in Part 1 directly inform your choices in Parts 2 through 6.

Read before starting

Questions marked [JUSTIFY] require a written response of at least one substantive paragraph grounded in statistical theory. These responses must reference the actual values, distributions, and context of your specific dataset. Generic answers that could apply to any dataset receive at most 40% of the available marks for that question.

PART 1 – Dataset Audit and Exploratory Analysis [15 marks]

Before any modelling begins, you must understand your data. This part is not a formality; the decisions you make here propagate through the entire analysis.

1.1 Structural Audit and Variable Classification [4 marks]

- (a) Produce a variable inventory table (one row per variable) containing: variable name, inferred dtype, correct type, measurement level (nominal/ordinal/discrete/continuous), and the role it plays in your primary question (predictor, outcome, confounder, identifier, or irrelevant).
- (b) Identify any variable whose measurement level is ambiguous. Write a paragraph defending your classification and explaining what analytical consequence a misclassification would have.
- (c) Compute a missing-data audit: count and percentage missing per variable. Produce a visualisation (a sorted bar chart or a heatmap) of missingness. For each variable with >5% missing, classify the mechanism as MCAR, MAR, or MNAR and justify your classification.
- (d) [JUSTIFY] For the two variables with the highest missingness, evaluate three handling strategies: listwise deletion, mean/median imputation, and multiple imputation (or a principled alternative). Implement at least two strategies, compare the resulting distributions numerically, and defend your final choice.

Dataset-specific guidance – Part 1.1

MIMIC: SpO2 and blood pressure fields have substantial missingness that is likely MNAR (sicker patients may not have recordings). Treating them as MCAR would be a critical error.

ACS PUMS: Income of zero is not missing; it represents genuinely zero earnings. Do not impute. Distinguish structural zeros from item non-response.

GTD: Missingness in weapon type correlates with attack type. Test this correlation before classifying the mechanism.

Stack Overflow: Compensation is missing for ~30% of respondents. Investigate whether missingness correlates with country or employment type.

WHO/WB: Missing health indicators strongly correlate with income group (MNAR). Imputing without acknowledging this introduces bias.

1.2 Univariate Distributions [4 marks]

- (a) For every continuous variable in your analysis, compute and tabulate: mean, median, SD, IQR, skewness coefficient, excess kurtosis, and the 5th and 95th percentiles.
- (b) Produce histograms for all continuous variables. Justify your bin count using the Freedman–Diaconis rule for at least three variables – show the computation.
- (c) [JUSTIFY] Your primary outcome variable (see Section 3) almost certainly requires a transformation. Diagnose its distribution formally using a Q-Q plot and the Shapiro-Wilk or Kolmogorov–Smirnov test. Apply the most appropriate transformation, reproduce the Q-Q plot, and compare skewness coefficients

before and after. Justify your choice of transformation with reference to the variable's data-generating process.

(d) For every categorical variable, compute frequency and proportion per category. Identify any category with fewer than 30 observations. Explain how rare categories will be handled before they enter a model (collapse, drop, or retain) and justify the decision.

1.3 Bivariate Relationships [4 marks]

(a) Compute the full Pearson correlation matrix for all continuous variables. Visualise as an annotated heatmap. Identify the three strongest pairwise correlations. For each, produce a scatter plot and comment on linearity, outlier influence, and whether Pearson is the right measure.

(b) For any pair of predictors with $|r| > 0.70$, compute Spearman's rank correlation and compare. [JUSTIFY] Explain the theoretical implication of high predictor-predictor correlation for regression modelling, referencing the concept of Variance Inflation Factor (VIF).

(c) Select two categorical predictors that are relevant to your primary question. For each, compare the distribution of the outcome variable across its categories using grouped box plots and a table of per-group statistics. Order groups by median, not by label.

(d) [JUSTIFY] Construct a 'data story' of 300–500 words that summarises your EDA findings and directly motivates the modelling choices you will make in Parts 2–5. The story must identify the most analytically important features, flag the two most serious data quality risks, and state your working hypothesis for the primary question.

1.4 Outlier Analysis [3 marks]

(a) Apply both the IQR fence rule and the Z-score method ($|z| > 3$) to your outcome variable and to the three most important continuous predictors. For each variable, report the count flagged by each method and the overlap.

(b) [JUSTIFY] For the outcome variable: inspect the five most extreme values by looking at the raw data. Are they plausible? Are they errors? Are they influential? State your decision for each (keep/remove/Winsorise) with a statistical and domain justification. Show how your decision changes the mean, SD, and skewness.

(c) Fit a preliminary simple linear regression of the outcome on the strongest predictor from your correlation matrix. Compute Cook's distance for each observation. Identify any observations with Cook's distance $> 4/n$. Refit the model with and without these points and compare the estimated slope, R^2 , and residual SD.

PART 2 – Inferential Statistics [15 marks]

Statistical inference is the process of drawing conclusions about a population from a sample. Every test you run must be motivated by your primary question or a clearly stated sub-question. Running tests without a prior question is p-hacking.

2.1 Estimation and Confidence Intervals [4 marks]

- (a) For your primary outcome variable, compute a 95% confidence interval for the population mean using: (i) the t-distribution formula (state and check the required conditions), (ii) a bootstrap percentile CI (10,000 resamples). Present both in a single summary table. Compare their widths.
- (b) [JUSTIFY] The t-based CI rests on specific assumptions. For each assumption, state it formally, describe the diagnostic check you used, and report the result of that check for your data. If any assumption is violated, explain whether you proceeded with the t-CI, why, and what the consequence is for the interpretation of the interval.
- (c) Compute a 95% bootstrap CI for the median of the outcome variable and for Spearman's rank correlation between the two variables with the highest $|r|$ from Part 1.3. [JUSTIFY] Explain why bootstrapping is preferred over parametric formulas for these two statistics.
- (d) Compute required sample sizes for the following scenario from your dataset: you wish to re-estimate the mean of the outcome variable to within ± 0.05 standard deviations at 95% confidence. Show the full derivation. Compare to the actual n you have and interpret the implication.

Dataset-specific guidance – Part 2.1

MIMIC: Compute the CI for mean SAPS-II score and for mean LOS. The LOS CI on raw data will be misleading, compute it on $\log(\text{LOS})$ and back-transform for interpretation.

ACS: Compute the CI for mean $\log(\text{wage income})$ for the full sample and separately by gender.

GTD: Compute the CI for mean number killed per incident. Note the extreme skew and zero-inflation.

Stack Overflow: Compute the CI for mean $\log(\text{compensation})$ separately for developers who use AI tools and those who do not.

WHO/WB: Compute the CI for mean U5MR separately for each World Bank income group.

2.2 Hypothesis Testing: One-Sample and Two-Sample Tests [5 marks]

- (a) Formulate three testable sub-questions motivated by your primary question. For each sub-question, state H_0 and H_1 , identify the appropriate test, state the conditions required for that test, check each condition, run the test, and report the test statistic, p-value, and decision at $\alpha = 0.05$.

Your three tests must span at least two of the following test families: one-sample t-test, two-sample Welch's t-test, paired t-test, Mann-Whitney U, chi-square test of independence.

- (b) [JUSTIFY] For at least one of your three tests, verify the result using a permutation test (10,000 permutations). Plot the null distribution with a vertical line at the observed test statistic. Compare the permutation p-value to the parametric p-value and explain any difference.
- (c) For each of your three tests that reach significance at $\alpha = 0.05$, compute an appropriate effect size measure: Cohen's d for continuous outcomes, Cramér's V for categorical associations. [JUSTIFY] For at least one test, show a case where a result is statistically significant but the effect size is negligible, and explain the practical implication.
- (d) [JUSTIFY] You have run three hypothesis tests. Apply the Bonferroni correction. Does any conclusion change? Explain what the family-wise error rate is, why it matters in the context of exploratory data analysis, and what the trade-off is between Type I and Type II error when you apply the correction.

2.3 Tests for Categorical Variables [3 marks]

- (a) Identify two pairs of categorical variables in your dataset that are substantively relevant to your primary question. For each pair, construct the contingency table of observed counts and run a chi-square test of independence. Report the test statistic, degrees of freedom, p-value, and Cramér's V.
- (b) Check the minimum expected cell count condition for each chi-square test. If the condition is violated, identify which cells are problematic and explain what you will do: collapse categories, use Fisher's exact test, or proceed with caution. Implement your chosen solution.
- (c) [JUSTIFY] For one of the two categorical associations you tested: break the contingency table down by a third variable (a potential confounder). Compute separate chi-square tests for each stratum. Do the results within strata agree with the aggregate result? If they do not, you may have found a Simpson's paradox. Describe exactly what is happening and what statistical/causal interpretation is correct.

2.4 Distribution Fitting [3 marks]

- (a) Select one continuous variable and one count variable from your dataset. For the continuous variable: fit a normal and a lognormal distribution by MLE. Overlay both fitted PDFs on a histogram. Produce Q-Q plots for both fits. Use the Kolmogorov–Smirnov test (`scipy.stats.kstest`) to formally compare fit quality and state which distribution better describes the data.
- (b) For the count variable: fit a Poisson distribution (estimate λ = sample mean). Compute the dispersion ratio (variance/mean). If over-dispersed (ratio > 1.5), also fit a Negative Binomial distribution. Overlay both PMFs on the empirical frequency distribution. [JUSTIFY] State the four Poisson conditions and assess each for your variable.
- (c) Demonstrate the Central Limit Theorem on your most skewed continuous variable: draw 5,000 bootstrap samples of sizes $n \in \{5, 15, 30, 100\}$. For each n , plot the distribution of sample means and overlay the theoretical CLT normal curve. Produce a 2×2 panel figure. [JUSTIFY] At which n does approximate normality hold for your specific variable? Does this match the $n \geq 30$ rule of thumb? Reference the skewness coefficient.

PART 3 – Regression Modelling [20 marks]

Regression is not about finding a single correct model. It is about understanding which variables explain variation in the outcome and by how much, subject to stated assumptions. Your choices here must be driven by your primary question.

3.1 Simple Linear Regression: Baseline Model [4 marks]

- (a) Identify the single continuous predictor most strongly associated with your outcome variable (use the correlation matrix from Part 1.3). Fit a SLR model: $\text{outcome} \sim \text{predictor}$ (both transformed as per Part 1.2c if applicable). Report: β_0 , β_1 , standard errors, t-statistics, p-values, R^2 , adjusted R^2 , and RMSE.
- (b) [JUSTIFY] Interpret β_1 precisely in the context of your dataset and primary question. If you transformed the outcome or predictor, interpret the coefficient on the original scale (e.g., a 1% increase in x is associated with a β_1 % change in y). Comment on whether the intercept is interpretable.
- (c) Compute and plot a 95% confidence interval for the slope using the t-distribution formula. Then compute a bootstrap 95% CI for the same slope (10,000 resamples). Compare the two intervals and explain any difference.
- (d) Plot residuals vs. fitted values and a Q-Q plot of residuals for this baseline model. [JUSTIFY] Interpret each plot in detail: identify any evidence of non-linearity, heteroskedasticity, or non-normality of residuals. State which LINE assumption(s) appear violated and what consequence that has for inference.

3.2 Multiple Linear Regression: Building the Full Model [8 marks]

- (a) Construct a sequence of four nested MLR models, each adding complexity: Model 1: $\text{outcome} \sim \text{strongest predictor}$ (baseline from 3.1). Model 2: add two more continuous predictors. Model 3: add the two most relevant categorical predictors (as dummies). Model 4: add at least one interaction term motivated by your primary question.
- (b) For each model report: adjusted R^2 , RMSE on training data, AIC, and BIC. Present in a single comparison table. [JUSTIFY] Explain why adjusted R^2 can decrease when a predictor is added, and why AIC and BIC may select different models.
- (c) For your preferred model (Model 3 or 4), interpret every coefficient in plain language, including the dummy variable reference category and any interaction term. For the interaction term: produce a visualisation that shows how the effect of one predictor changes as a function of the other (e.g., a set of regression lines for different levels of the moderating variable).
- (d) Compute VIF for all predictors in your preferred model. Identify any predictor with $\text{VIF} > 5$. [JUSTIFY] Explain what multicollinearity does to the standard errors of the affected coefficients and why this is a problem for inference, even if the overall model fit is good. If $\text{VIF} > 10$ for any predictor, implement a remedy (remove, combine, or defer to regularisation in Part 5) and refit.
- (e) Run the full diagnostic suite on your preferred model: (i) residuals vs. fitted values, (ii) Q-Q plot of residuals, (iii) scale-location plot ($\sqrt{|\text{residuals}|}$ vs. fitted), (iv) residuals vs. leverage with Cook's distance contours. For each plot, state what you are looking for and what you see.
- (f) [JUSTIFY] If the residual plots indicate heteroskedasticity, apply a variance-stabilising transformation or discuss the use of heteroskedasticity-robust standard errors (HC3). Show how the coefficient standard errors change and what that implies for the significance of individual predictors.

3.3 Feature Engineering and Model Refinement [4 marks]

- (a) Identify one predictor where the relationship with the outcome appears non-linear (use residual plots or scatter plots from Part 1.3). Add a polynomial term (x^2) or apply a log transformation to that predictor. Refit the model, compare AIC and the residual vs. fitted plot before and after, and justify whether the added complexity is warranted.

(b) Add a new engineered feature of your own design that is motivated by domain knowledge about your dataset. Explain what the feature represents, why you expect it to be a useful predictor, and show whether adding it improves AIC and adjusted R^2 .

(c) [JUSTIFY] Write a 250–400 word “Modelling Choices” section in your report that explains, for every decision you made in Part 3 (variable selection, transformations, interaction terms, outlier treatment), the statistical and domain reasoning behind the decision. Acknowledge at least two decisions where a reasonable analyst might have chosen differently and explain the trade-off.

3.4 Inference on Coefficients [4 marks]

(a) For your final MLR model, report the 95% confidence interval for each coefficient. Identify which predictors have CIs that exclude zero and which do not. [JUSTIFY] Explain the relationship between the t-test for a single coefficient and the F-test for the overall model. Under what circumstance can the F-test be significant while all individual t-tests are not?

(b) Compute a prediction interval for a new, hypothetical observation whose predictor values you specify (choose values that are within the observed data range). Report both the 95% confidence interval for the mean response and the 95% prediction interval for the individual response at that point. [JUSTIFY] Explain why the prediction interval is always wider than the confidence interval for the mean response.

(c) Using your final MLR model, calculate the partial R^2 for the predictor that is most central to your primary question (i.e., the ΔR^2 from adding it to a model that already contains all other predictors). Interpret this as a measure of unique explanatory power.

Dataset-specific guidance – Part 3

MIMIC: Outcome = $\log(\text{LOS})$. Key predictors: SAPS-II, admission type, number of ICU stays, age. Interaction to test: SAPS-II $\times$ admission type.

ACS PUMS: Outcome = $\log(\text{wage income})$. Key predictors: education, experience (age–education–22), hours worked, sex, race. Non-linearity: experience likely has diminishing returns – test quadratic term.

GTD: Outcome = $\log(1 + \text{number killed})$. Key predictors: attack type, weapon type, region, suicide indicator. Interaction: weapon type $\times$ suicide indicator.

Stack Overflow: Outcome = $\log(\text{compensation})$. Key predictors: experience, education, AI tool use, remote status, developer type, region. Interaction: experience $\times$ AI tool use. –

WHO/WB: Outcome = $\log(\text{U5MR})$. Key predictors: $\log(\text{health expenditure})$, physician density, vaccination coverage, $\log(\text{GDP})$, income group. Interaction: $\log(\text{health expenditure}) \times \text{income group}$.

PART 4 – Classification and Logistic Regression [15 marks]

Logistic regression models the probability of a binary outcome. You must define your binary outcome carefully and justify the classification threshold you use.

4.1 Binary Outcome Definition and Baseline [3 marks]

- (a) Define a binary outcome variable for your dataset (either use an existing binary variable or dichotomise a continuous/count variable at a substantive threshold). [JUSTIFY] Your justification must state the threshold, the proportion of positives it produces, and why this threshold is analytically meaningful for your primary question (not just convenient).
- (b) Report the class balance. If the minority class is below 20% of the sample, discuss the implications for classifier evaluation metrics and for the default 0.5 decision threshold.

Dataset-specific binary outcomes

MIMIC: Use the existing 28-day in-hospital mortality indicator (~11% positive).

ACS PUMS: Define high earner = 1 if wage income > median for their occupation group; 0 otherwise.

GTD: Use the existing success indicator (attack was 'successful' per GTD definition).

Stack Overflow: Define high compensation=1 if log(compensation) > 75th percentile for their region; 0 otherwise

WHO/WB: Define high-burden country-year=1 if U5MR > median for that income group in that year; 0 otherwise

4.2 Logistic Regression Modelling [6 marks]

- (a) Fit two logistic regression models using `statsmodels.formula.api.logit()`: Model L1: binary outcome ~ three most important predictors from your MLR analysis. Model L2: binary outcome ~ all predictors from your best MLR model (adapted for logistic). For each model report: coefficients, standard errors, z-statistics, p-values, odds ratios with 95% CIs, and McFadden's pseudo- R^2 .
- (b) [JUSTIFY] For three predictors in Model L2: interpret the odds ratio precisely (e.g., "a one-unit increase in X multiplies the odds of outcome = 1 by $\exp(\beta)$ "). For a continuous predictor, also interpret the odds ratio for a one-standard-deviation increase. For a categorical predictor, state clearly what the reference category is and what comparison the odds ratio captures.
- (c) Compare Model L1 and L2 using AIC and a likelihood ratio test (LRT). [JUSTIFY] Explain what the LRT is testing and how to interpret its p-value. State which model you prefer and why, acknowledging the parsimony-fit trade-off.
- (d) Check the logistic regression assumptions: (i) linearity of log-odds – use the Box-Tidwell test or a visual approach for each continuous predictor; (ii) absence of perfect separation; (iii) absence of strongly influential observations (use standardised Pearson residuals and leverage). [JUSTIFY] For any violated assumption, explain the consequence for the model estimates.

4.3 Model Evaluation [6 marks]

- (a) Using a stratified 80/20 train/test split, evaluate your preferred logistic model on the test set. Report: accuracy, precision, recall, F1-score, and AUC. Present the confusion matrix as a heatmap.
- (b) [JUSTIFY] The default classification threshold of 0.5 is rarely optimal for imbalanced classes. Plot the precision-recall curve and the ROC curve for your model. Identify a threshold that maximises the F1-score. Re-evaluate accuracy, precision, recall, and F1 at this optimal threshold. Explain what trade-off you are making relative to the default.
- (c) [JUSTIFY] AUC measures discrimination. Explain what AUC = 0.5 means, what AUC = 1.0 means, and what your observed AUC tells you about your model's discriminative ability in the context of the primary question. Discuss: is high AUC sufficient evidence that the model is useful? What other criteria matter in your specific domain?
- (d) Compute calibration: bin predicted probabilities into 10 equal-width bins and compare mean predicted probability to observed proportion of positives within each bin. Plot the calibration curve (reliability diagram). Is the model well-calibrated, overconfident, or underconfident?

PART 5 – Model Validation and Regularisation [15 marks]

A model evaluated only on its training data is not evaluated at all. This part requires you to implement robust validation and explore regularisation as a tool for improving generalisation.

5.1 Cross-Validation [6 marks]

- (a) Implement 5-fold cross-validation for your preferred MLR model from Part 3. Report: the CV-RMSE for each fold, the mean CV-RMSE, and the standard deviation across folds. Compare the training RMSE to the CV-RMSE. [JUSTIFY] What does a large discrepancy between training RMSE and CV-RMSE indicate?
- (b) Repeat 5-fold CV for all four nested models from Part 3.2a. Plot CV-RMSE $\pm$ 1 standard deviation as a function of model complexity. [JUSTIFY] Does the CV-RMSE curve show a clear minimum, or does it plateau? What does this tell you about overfitting in your preferred model?
- (c) Implement 5-fold CV for your preferred logistic model from Part 4 and report CV-AUC (mean and SD across folds) and CV-F1 (at the optimal threshold from Part 4.3b). [JUSTIFY] Why is stratified k-fold CV important for imbalanced binary classification problems?
- (d) [JUSTIFY] Explain the bias-variance trade-off formally. Sketch (or produce computationally) the expected U-shaped test error curve as a function of model flexibility. Identify where in this curve you believe your preferred MLR model sits, with reference to your CV results.

5.2 Ridge and Lasso Regularisation [6 marks]

- (a) Using sklearn, fit a Ridge regression and a Lasso regression on your full predictor matrix (standardise all predictors before fitting). Use a 5-fold CV to select the optimal λ for each method (use sklearn RidgeCV and LassoCV or GridSearchCV). Report the optimal λ and the resulting CV-RMSE for each.
- (b) Produce a coefficient path plot for Lasso: plot the coefficient of each predictor as a function of $\log(\lambda)$. Mark the optimal λ with a vertical line. [JUSTIFY] Which predictors are driven to exactly zero at the optimal λ ? Does this variable selection agree with the predictors that were statistically significant in your OLS MLR model? Discuss any discrepancies.
- (c) Produce a comparison table of OLS, Ridge, and Lasso using three metrics: training RMSE, 5-fold CV-RMSE, and number of non-zero coefficients. [JUSTIFY] Explain theoretically why Ridge retains all predictors while Lasso zeros some out, referencing the geometry of L1 vs. L2 penalty.
- (d) [JUSTIFY] Your dataset almost certainly has multicollinear predictors (flagged by VIF > 5 in Part 3.2d). Explain what Ridge regression does to the coefficients of correlated predictors compared to OLS, and why this is often a more stable solution than manually removing one of the correlated variables. Demonstrate this by comparing the Ridge coefficients for the collinear pair to their OLS counterparts.

5.3 Final Model Selection [3 marks]

- (a) Produce a final model selection summary table comparing: OLS (best MLR), Ridge, Lasso, and Logistic. For each model report: CV-RMSE or CV-AUC, as appropriate; number of predictors; interpretability (score 1–5); and suitability for the primary question (score 1–5). State clearly which model you recommend and why.
- (b) [JUSTIFY] Write a “Model Recommendation” paragraph of 200–300 words addressed to a non-technical decision-maker. State your recommended model, what it predicts and how well, what its key predictors are, and what its most important limitations are. Avoid statistical jargon – be precise but accessible.

PART 6 – Critical Synthesis and Statistical Pitfalls [10 marks]

Every real analysis contains decisions that could have been made differently, and conclusions that are weaker than they first appear. This part requires you to stress-test your own work.

6.1 Simpson’s Paradox Investigation [4 marks]

- (a) Identify one association in your data that could be subject to Simpson’s paradox: a relationship between two variables that changes direction or magnitude when a third variable is controlled for. Compute the aggregate association and then stratify by the third variable. Show the aggregate and stratified results side-by-side using both numerical summaries and plots.
- (b) [JUSTIFY] Explain exactly why Simpson’s paradox arises in your specific case: what is the confounder, how is it distributed across groups, and why does controlling for it reverse or attenuate the association? What is the correct interpretation – the aggregate or the stratified result – and why?

Dataset-specific hints – Simpson’s Paradox

MIMIC: Aggregate mortality rate by gender; then stratify by admission type or age group. Older patients may be more likely to be female AND more likely to die, reversing the crude gender-mortality association.

ACS PUMS: Aggregate mean wage by education; then stratify by occupation. Within occupations, education may show a smaller or reversed effect due to occupational sorting.

GTD: Aggregate success rate by region; then stratify by attack type. Some regions specialise in attack types with higher baseline success, confounding the regional comparison.

Stack Overflow: Aggregate mean compensation by AI tool usage; then stratify by years of experience. Experienced developers both earn more and are more likely to use AI tools.

WHO/WB: Aggregate U5MR vs. health expenditure across all countries; then stratify by income group. Within income groups, the relationship may be weaker or reversed due to baseline structural differences.

6.2 Threats to Validity [3 marks]

- (a) [JUSTIFY] Identify and discuss three specific threats to the internal or external validity of your analysis. For each threat, state: what it is, why it applies to your dataset specifically, what its likely direction of effect is on your key estimates, and what you could do to mitigate it with more data or a different study design.
- (b) Your analysis involves multiple hypothesis tests across Parts 2, 3, and 4. Count the total number of significance tests you ran. [JUSTIFY] Compute the expected number of false positives under the null at $\alpha = 0.05$. Identify two findings in your report that you now regard as potentially spurious due to multiple testing, and explain what additional evidence would be needed to confirm them.

6.3 What the Model Cannot Tell You [3 marks]

- (a) [JUSTIFY] Your regression model is correlational, not causal. Identify the three most important unmeasured confounders that are absent from your dataset but almost certainly influence both your key predictor and your outcome. For each, state the likely direction of the omitted variable bias it introduces.
- (b) Describe the population to which your model can be generalised. Is the dataset a random sample, a convenience sample, or a census? Identify at least one population that your model should NOT be applied to, and explain the statistical and domain reasoning.
- (c) [JUSTIFY] Write a “Limitations” section of 200–350 words for your technical report. It must cover: data quality limitations, modelling assumption violations, threats to validity, and scope of generalisability. It must be specific to your dataset and findings – not a generic paragraph about statistics.

6. Presentation [10 marks]

Each group presents their analysis in Week 13 (15 minutes + 5 minutes Q&A). The Q&A is not optional – all group members may be questioned individually on any aspect of the project.

Criterion	Description	Marks
Clarity of Problem Statement	The primary question is clearly stated and the analysis is structured around it.	2
Depth of Methodology	Modelling choices are justified, not just described. Assumptions are acknowledged.	3
Integrity of Conclusions	Conclusions are supported by evidence and limitations are honestly stated.	2
Q&A Performance	Members answer follow-up questions accurately, demonstrating individual understanding.	3

Q&A policy

The instructor may ask any group member to explain any element of the project – including code, derivations, and interpretations. A member who cannot answer questions about their own sections may have their individual mark adjusted downward even if the group report is strong. The Group Contribution Statement is reviewed before the Q&A.

7. Assessment Rubric

Each question is marked on four dimensions. The rubric below applies uniformly across all parts.

Dimension	Excellent (90–100%)	Proficient (75–89%)	Developing (55–74%)	Insufficient (<55%)
Statistical Correctness	Methods applied correctly with no errors. Results are reproducible.	Minor computational errors that do not affect conclusions.	Errors present that affect some conclusions but not all.	Fundamental errors in application of methods.
Justification Depth	Every decision grounded in statistical theory. Assumptions explicitly checked against the data.	Most decisions justified. Some assumptions stated but not checked.	Some justification present but generic or not connected to actual data values.	Decisions made without justification; assumptions ignored.
Interpretation Quality	Precise, dataset-specific interpretation. Uncertainty acknowledged. Practical significance discussed.	Mostly correct interpretation with minor imprecision.	Superficial interpretation; some errors (e.g., confusing statistical and practical significance).	No interpretation or substantially incorrect (e.g., correlation treated as causation).
Code & Reproducibility	Clean, documented, modular. Re-runs without error. Meaningful variable names.	Mostly clean. Minor undocumented steps. Re-runs successfully.	Some undocumented or poorly structured sections. Re-runs with minor fixes.	Does not re-run. Undocumented. Output inconsistent with code.

What distinguishes excellent from proficient

An excellent response to a JUSTIFY question references the specific numerical values in the dataset (e.g., "The VIF for physician density is 8.3, indicating substantial multicollinearity with $\log(\text{GDP})$. This inflates the standard error of the physician density coefficient by a factor of approximately $\sqrt{8.3} = 2.9$, making it appear non-significant at $\alpha = 0.05$ when it may genuinely predict U5MR."). A proficient response states the concept correctly but does not connect it to the actual values. A developing response states the concept but not its implication. An insufficient response omits the concept entirely.

8. Academic Integrity and Conduct

All members of a group are jointly and equally responsible for the submitted work. The Group Contribution Statement is a formal document – falsifying it constitutes academic misconduct.

You may use AI coding assistants (e.g., GitHub Copilot, ChatGPT) to assist with code. You may not submit AI-generated written justifications as your own work. The JUSTIFY responses must reflect your own statistical reasoning applied to your specific dataset. Responses that are factually incorrect, generic, or disconnected from your data values will be penalised regardless of their source.

All data sources must be cited. If you use a derived or pre-processed version of a dataset, cite both the original source and describe your processing steps in an appendix.

Plagiarism between groups

Groups may discuss general methodological approaches but may not share code, analysis outputs, or written text. Submissions with substantial similarity in code structure, results, or written responses will be referred for investigation. Submitting the same or similar analysis as another group, even on a different dataset, is not permitted.